Goursat's Theorem

> **Theorem 0.0.3.** Let G be an open set and let $f : G \to \mathbb{C}$ be a holomorphic function.
> Then,
>
> $$\int_T f = 0$$
>
> for every triangular path T in G.

Proof. Let $T_0 = [a, b, c, a]$ be the triangular path in G, and let $d = \operatorname{diam} T$ and $L = \operatorname{length} T$.
And, construct four subtriangles

$$T^{(1)} = \left[a, \frac{a+b}{2}, \frac{a+c}{2}, a\right],\ T^{(2)} = \left[\frac{a+b}{2}, b, \frac{b+c}{2}, \frac{a+b}{2}\right],\ T^{(3)} = \left[\frac{a+c}{2}, \frac{b+c}{2}, c, \frac{a+c}{2}\right],\ T^{(4)} = \left[\frac{a+b}{2}, \frac{b+c}{2}, \frac{a+c}{2}, \frac{a+b}{2}\right]$$

Then,

$$\int_{T_0} f = \int_{T_1} f + \int_{T_2} f + \int_{T_3} f + \int_{T_4} f$$

Thus, by the triangle inequailty,

$$\left|\int_{T_0} f\right| \leq \left|\int_{T_1} f + \int_{T_2} f + \int_{T_3} f + \int_{T_4} f\right| \leq 4 \cdot \max_{i=1,2,3,4} \left|\int_{T_i} f\right|$$

Let $T_1 = T^j$, where the j maximizes the above integral. Apply the same process for $T_0, T_1, \ldots, T_{n-1}$, we obtain that a sequence of nested compact sets $\{\mathcal{T}_n\}$, where $\mathcal{T}_n$ is the convex hull of T_n. So, the intersection

$$\bigcap_{n=0}^{\infty} \mathcal{T}_n = \{z_0\}$$

is a singleton. Given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|z - z_0| < \delta \implies \left|\frac{f(z) - f(z_0)}{z - z_0} - f'(z_0)\right| < \varepsilon$$

since f is holomorphic. Moreover, by the construction, there exists $N \in \mathbb{N}$ such that

$$T_N \subseteq B(z_0; \delta)$$

Now, the triangle inequality again,

$$\left|\int_{T_N} f(z)dz\right| = \left|\int_{T_N} f(z) - f(z_0) - f'(z_0)(z - z_0)dz\right| \leq \varepsilon \int_{T_N} |z - z_0||dz| = \varepsilon \cdot \operatorname{diam} T_N \cdot \operatorname{length} T_N$$

Finally, since $\operatorname{diam} T_N = 2^{-n} \operatorname{diam} T_0$ and $\operatorname{length} T_N = 2^{-n} \operatorname{length} T_0$,

$$\left|\int_{T_0} f(z)dz\right| \leq 4^N \cdot \left|\int_{T_N} f(z)dz\right| \leq 4^N \cdot \varepsilon \cdot \operatorname{diam} T_N \cdot \operatorname{length} T_N \leq \varepsilon \cdot \operatorname{diam} T_0 \cdot \operatorname{length} T_0 \to 0 \ \text{ as } \ \varepsilon \to 0$$

□

「Thm. Let G be an open set of $\mathbb{C}$, and T be a triangluluar path contained in G. For a point a inside in T, suppose that f is Holomorphic on $G \setminus \{a\}$. If f is bounded on for some neighborhood of a, then

$$\int_T f = 0.$$

Proof. For fixed a, construct a subtriangle T^1, T^2, T^3, T^4 from T. Then, for some T^k, $a \in T^k$. (There are at most three case). Put this $T_1 = T^k$. Then, inductively, we can construct $\{T_n\}$ compact nested, and

$$\left|\int_T f\right| \leq \left|\sum_{a \in T_n^k} \int_{T_n^k} f\right| \leq \frac{3L}{2^n} \cdot \sup|f| \to 0 \text{ as } n \to 0.$$

「Morera Theorem.

Let $f : D \to \mathbb{C}$ be a continuous function on an open set D. Suppose that for every triangle T inside G,

$$\int_T f = 0.$$

Then, f is Holomorphic on G.

Proof. Let $z_0 \in G$ be fixed, and $B_r(z_0)$ be given where r was chosen by $B_r(z_0) \subseteq G$. For each $z \in B_r(z_0)$, define

$$F(z) = \int_{[z_0, z]} f(u)\, du \text{ where } [z_0, z] \text{ is the straight segment.}$$

For small $h \in \mathbb{C}$, observe that

$$\frac{F(z+h) - F(z)}{h} = \frac{1}{h} \cdot \int_{[z, z+h]} f(u)\, du, \text{ because}$$

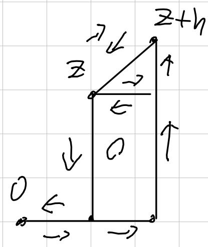
$$\int_{[z_0, z]} f + \int_{[z, z+h]} f + \int_{[z+h, z_0]} f = 0, \text{ by the assumption.}$$

Now, the continuity gives $\lim_{h \to 0} \int_{[z, z+h]} f(u)\, du = f(z)$. ☒

Thm. A holomorphic function $f: D \to \mathbb{C}$ has a primitive where D is open disk.

Proof. WLOG, let $D = B(0; R)$. Given $z = a + bi \in D$, define

$$F(z) = \int_{\gamma_z} f(s)\, ds \qquad \text{where } \gamma_z = [(0,0), (a,0), (a,b)]$$



Now, let $h \in \mathbb{C}$ with $0 < |h| < R - |z|$. Then,

$$F(z+h) - F(z) = \int_{\gamma_{z+h}} f(s)\, ds - \int_{\gamma_z} f(s)\, ds = \int_{[z, z+h]} f(s)\, ds$$

using Goursat Thm in Here, with holomorphic.

Given $\varepsilon > 0$, there is $\delta > 0$ such that

$$|z - s| < \delta \Rightarrow |f(z) - f(s)| < \varepsilon.$$

So,

$$\left| \frac{1}{h} \cdot \int_{[z,z+h]} f(s)\, ds - f(z) \right| = \left| \frac{1}{h} \cdot \int_{[z,z+h]} f(s) - f(z)\, ds \right| < \varepsilon, \text{ for small enough } h. \quad \blacksquare$$

Corollary. If f is Homolophic on an open disk, then for any closed curve γ in the disk,

$$\int_\gamma f = 0.$$

「Cauchy integral formula

「Thm. Let $D = \{z \in \mathbb{C} \mid |z - z_0| \leq r\}$ be a disk centered at z_0, and $C : t \mapsto z_0 + r \cdot e^{it}$ be a curve, along the boundary of D. Suppose that $f : G \to \mathbb{C}$ is a holomorphic on an open G containing D. Given $z \in D$,

$$f(z) = \frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{\mu - z} d\mu$$

Proof. Let $z \in D$ be fixed. Since $f(\mu)/\mu - z$ is holomorphic except $\mu = z$, for a cure $C_\varepsilon : t \mapsto z + \varepsilon \cdot e^{it}$ where $\varepsilon < r$, we obtain

$$\int_{-C_\varepsilon} \frac{f(\mu)}{\mu - z} d\mu + \int_C \frac{f(\mu)}{\mu - z} d\mu = 0, \quad \cdots (*)$$

by the Cauchy-Goursat theorem.

Since f is holomorphic, the derivative of f near z is bounded, so

$$\int_{C_\varepsilon} \frac{f(\mu) - f(z)}{\mu - z} d\mu \to 0 \quad \text{as } \varepsilon \to 0.$$

and, for any $\varepsilon > 0$,

$$\int_{C_\varepsilon} \frac{f(z)}{\mu - z} d\mu = f(z) \cdot \int_0^{2\pi} \frac{1}{\varepsilon \cdot e^{it}} \cdot i \cdot \varepsilon e^{it} dt = f(z) \cdot 2\pi i.$$

Therefore,

$$\int_{-C_\varepsilon} \frac{f(\mu)}{\mu - z} d\mu = -\int_{C_\varepsilon} \frac{f(\mu) - f(z)}{\mu - z} d\mu - \int_{C_\varepsilon} \frac{f(z)}{\mu - z} d\mu \to -2\pi i \cdot f(z),$$

as $\varepsilon \to 0$, so we obtain the result, taking $\varepsilon \to 0$ to $(*)$. ◻

Thm. Suppose f is Holomorphic on an open G containing a closed disk D, having center z_0,

$$f(z)=\sum_{n=0}^{\infty}\frac{f^{(n)}(z_0)}{n!}\cdot(z-z_0)^n \qquad (z\in D)$$

Proof. $f(z)=\frac{1}{2\pi i}\cdot\int_C\frac{f(\mu)}{\mu-z}\,d\mu \qquad \left(\frac{1}{\mu-z}=\frac{1}{\mu-z_0-(z-z_0)}=\frac{1}{\mu-z_0}\cdot\frac{1}{1-\left(\frac{z-z_0}{\mu-z_0}\right)}\right)$

$$=\frac{1}{2\pi i}\cdot\int_C f(\mu)\cdot\sum_{n=0}^{\infty}\frac{1}{(\mu-z_0)^{n+1}}\cdot(z-z_0)^n\,d\mu$$

$$=\sum_{n=0}^{\infty}\left(\frac{1}{2\pi i}\cdot\int_C\frac{f(\mu)}{(\mu-z_0)^{n+1}}\,d\mu\right)\cdot(z-z_0)^n$$

$$=\sum_{n=0}^{\infty}\frac{f^{(n)}(z_0)}{n!}\cdot(z-z_0)^n.$$

Corollary, Liouville's Theorem.

If $f:\mathbb{C}\to\mathbb{C}$ is entire and bounded, then it is constant.

Proof. Given $z_0\in\mathbb{C}$ and $R>0$, $|f'(z_0)|\leq\frac{1}{R}\cdot\sup_{C_R}|f|\to 0$ as $R\to\infty$. Therefore, $f'=0$

Thus f is constant.

Fundamental Theorem of Algebra.

Non-constant polynomial $P(z)=a_n z^n+a_{n-1}z^{n-1}+\cdots+a_0\in\mathbb{C}[z]$ has a root in $\mathbb{C}$.

Proof. Suppose that $P(z)$ has no root in $\mathbb{C}$. Then, $1/P(z)$ is well-defined on $\mathbb{C}$, entire.

observe that $\frac{P(z)}{z^n}=a_n+\left(\frac{a_{n-1}}{z}+\cdots+\frac{a_0}{z^n}\right) \quad (z\neq 0).$

The terms $a_{n-k}\cdot z^{-k}$ $(k=1,\cdots,n)$ converges to zero as $|z|\to\infty$, thus for some $R>0$,

$$|P(z)|\geq 2|a_n|\cdot z^n \quad (|z|>R)$$
$$\geq 2|a_n|\cdot R^n$$

Therefore it is bounded below in $|z|>R$, and clearly in $|z|\leq R$.

Now, $\frac{1}{P}$ is bounded, thus constant. Contradiction.